

Ananya Rajan

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SUMMARY

Final-year Computer Science student with hands-on experience building and shipping machine learning models, from churn prediction to a GenAI document Q&A agent. Comfortable moving between statistical rigor (hypothesis testing, A/B design) and production code (Python, SQL, Scikit-learn). Learned most of this by building three projects end to end and defending the choices behind each one.

EDUCATION

B.Tech Computer Science and Engineering | CGPA 8.4 | 2026

Vellore Institute of Technology, Vellore

Class 12 (CBSE): 92%, 2022 | Class 10 (CBSE): 95%, 2020 | No active backlogs

EXPERIENCE

Data Science Intern - Loopwise Analytics | Bengaluru, India | May 2025 - Jul 2025

- Built a churn prediction model on 3 years of telecom subscriber data using XGBoost, reaching 84% recall on the held-out test set.
- Chose SMOTE oversampling over class weighting after testing both, since the churn class was under 8% of the data and class weighting alone missed too many true churners.
- Presented findings to the client's retention team with a ranked list of the 500 highest-risk accounts, not just a model accuracy number.

PROJECTS

Document Q&A Agent - Answers questions over internal PDFs and wikis *Python, LangChain, OpenAI API, FAISS, Streamlit*

- Built a retrieval-augmented Q&A agent over 200+ internal documents, chunking by heading structure instead of fixed token windows after fixed windows kept splitting answers mid-table.
- Cut average query latency from 6s to 2.1s by caching embeddings instead of recomputing them per session.

A/B Test Analysis Dashboard - Statistical significance checker for marketing experiments *Python, Pandas, SciPy, Plotly*

- Built a dashboard that runs two-proportion z-tests and reports required sample size before a test launches, after a prior campaign was called early without enough samples to be conclusive.
- Used sequential testing correction instead of a single fixed-horizon test, so marketers checking results mid-flight would not inflate false positives.

SKILLS

Languages: Python, SQL

ML/Stats: Scikit-learn, XGBoost, hypothesis testing, A/B testing, regression, classification, clustering

Tools: Pandas, NumPy, LangChain, Streamlit, Git

Familiar with: PyTorch (built one CNN for a coursework assignment, not yet independent)